Correlation and Simple Regression

AGRON 5130 - Unit 10 Assignment

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Introduction

The goal of this assignment is to help you practice the full workflow from Module 10:

- visualizing quantitative relationships
- interpreting correlation carefully
- fitting and interpreting a simple linear regression
- checking model assumptions
- making predictions and distinguishing interpolation from extrapolation

You will use a wheat–weed dataset collected in India. The key biological question is whether increasing weed pressure is associated with lower wheat yield.

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

```r
library(ggplot2)

weed <- read.csv("data/wheat_weed_india.csv")
head(weed)
```

```
##      yield total_weed_pop total_weed_dm n_removal_weed n_uptake_wheat
## 1 4.236967       16.87627      17.27988       5.385811       3.537604
## 2 1.655525       20.31061      17.59948       5.694649       3.366266
## 3 6.529892       15.56713      16.80601       4.979389       3.719664
## 4 5.526334       17.41575      18.07812       5.135030       3.622897
## 5 9.061101       17.94017      14.85238       4.596280       4.029803
## 6 5.843537       16.54820      16.64260       5.076142       3.681508
```

Question 1

1 point

Create a scatterplot matrix for all quantitative variables in `weed`.

Question 2

1 point

Create a correlation matrix for all quantitative variables in `weed`. Briefly identify which pair of variables shows the strongest association.

Question 3

1 point

Create a scatterplot showing the relationship between `total_weed_pop` and `yield`.

Optional but recommended: overlay a linear trend line.

Question 4

1.5 points

Run a correlation test between `total_weed_pop` and `yield` using `cor.test()`.

In 2–3 sentences, interpret:

1. direction of association
2. strength of association
3. whether this result implies causation

Question 5

1.5 points

Fit a simple linear regression model for `yield` as a function of `total_weed_pop`.

Display coefficient estimates using `coef()`.

Question 6

1 point

Display R^2 for the model using `summary()`.

Question 7

2 points

Interpret the model from Questions 5–6.

Address all of the following:

1. Is the slope positive or negative?
2. Is the slope significantly different from zero?
3. What percentage of yield variation is explained by `total_weed_pop`?
4. In plain language, what does this model imply biologically?

Question 8

1.5 points

Check model assumptions for the regression from Question 5.

Produce:

1. residuals vs fitted plot (`which = 1`)
2. normal Q-Q plot (`which = 2`)

Then write 2–3 sentences on whether a straight-line model appears adequate.

Question 9

1 points

Using `ggplot()`, create a plot of `yield` vs `total_weed_pop` with:

1. observed points
2. fitted regression line

Question 10

2 points

Using the `new_weed_pops` data frame below, predict yield for each value of `total_weed_pop`.

Note: negative values are included intentionally to explore extrapolation.

```
new_weed_pops <- data.frame(
  total_weed_pop = c(-1, 1, 3, 5, 7)
)
```

Generate:

1. predicted values
2. 95% confidence intervals for the mean response

Hint: use `predict(..., interval = "confidence")`

Question 11

1.5 points

From Question 10, classify each prediction as:

1. interpolation
2. extrapolation

Justify your classification using the observed range of `total_weed_pop` in the original data.

Which predictions would you trust the most? Why?

Optional Bonus Question

+1 point

Based on your residual plots, do you see any signs that the assumptions of the linear model may be violated? If so, what alternative approach might be more appropriate?